

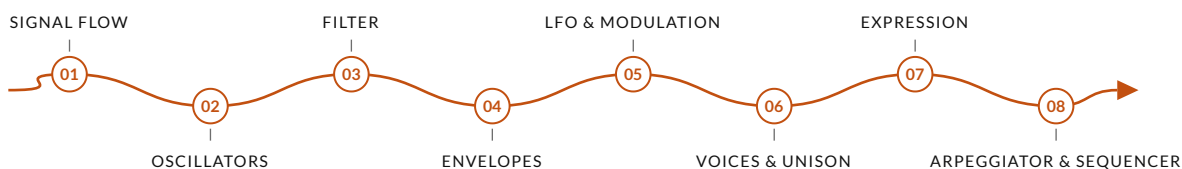


STUDENT HANDBOOK

INTRO TO SYNTHESIS

A crash course in subtractive synthesis, on the Sequential Fourm. Everything from the workshop in one place: the theory, the diagrams, and the words.

A COMPANION HANDBOOK · WRITTEN BY IULIAN CAZAN
SINTEZAUR × ZEEDO · POWERED BY SEQUENTIAL



SINTEZAUR

8 MODULES

ZEEDO

SEQUENTIAL

BEFORE WE START

Set MAIN, the volume knob, to about 12 o'clock and leave it there: from here on the sound changes with the other knobs, not with the volume.

Home base is the Basic Program: a plain sawtooth with the filter open and a vibrato already wired to the mod wheel (we use it in module 05). It doesn't matter if your sound doesn't come out identical to the recipe, that is the good part of exploring. If you get lost, come back to the Basic Program: hold TRANSPOSE DOWN and press PROGRAM. One thing to avoid: when the display shows a program name, turning SELECT loads another program and the sound you built is gone. Turn it only when a step asks you to, and when the display shows a parameter instead of a program name.

This handbook follows the workshop stop by stop. Each module has one page: the idea in a few lines, the diagram from the screen, and a TRY IT you can do at the synth. The words we use are collected in the glossary at the back. Read one module, turn one knob, listen.

WHAT THE BASIC PROGRAM IS

Osc A on sawtooth, full up in the mixer; on Osc B the sawtooth is lit too, but its mixer level sits at zero, so you hear nothing from it until you raise it. Filter open, no resonance, KEY TRACK off; both envelopes plain, VELOCITY off; feedback and noise off; in MODULATION the LFO is already routed to FREQ A and FREQ B on the mod-wheel bus, so raise the wheel and you have vibrato; the arpeggiator off. The same program fills the whole of User bank 2, and it is what TRANSPOSE DOWN + PROGRAM brings back.

SIX PAIRS OF WORDS

When you describe a sound, reach for these before anything else: bright / dark (how much high end) · thin / fat (how much body) · hollow / full (square-like or saw-like) · soft / hard (how it starts) · pluck / pad (does it die or does it stay) · static / moving (does anything change while it sounds).

Say which one you hear, then look for the knob that moves it: brightness lives at CUTOFF, body at the sub and the detune, hollowness at the waveshape, the start at ATTACK, the pluck at DECAY and SUSTAIN, the movement at the LFO.

Sequential's Prophet-5, in 1978, was the first fully programmable polysynth: before it a patch lived on a sheet of paper, rebuilt knob by knob.

SIGNAL FLOW

Every sound a subtractive synthesizer makes travels the same road: an oscillator generates a raw, harmonically rich tone, the mixer balances the sources, the filter sculpts the brightness, and the amplifier shapes the loudness over time. Learn this path and any knob you touch has an obvious home: you know which stage it belongs to, so you know what it does. Fourm is our instrument for the journey.

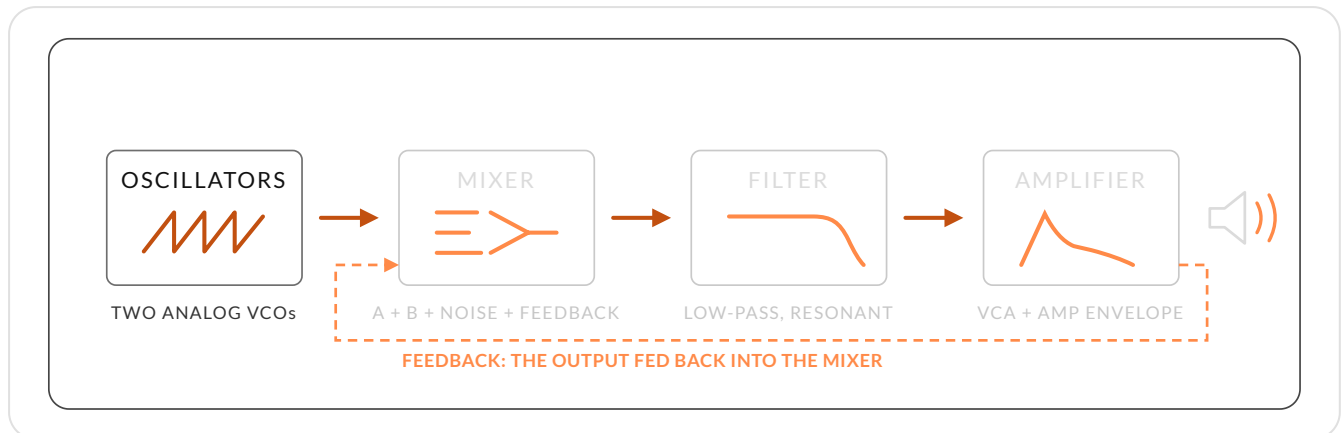


FIG. 1 · THE VOICE PATH: OSCILLATORS → MIXER → FILTER → AMPLIFIER → OUTPUT, WITH THE FEEDBACK RETURN.

WHAT'S IN A SOUND

A note is a fundamental frequency plus a stack of quieter overtones above it: the harmonics. Their mix is what you hear as timbre. A sawtooth carries every harmonic, a square only the odd ones, a triangle the odd ones too, but much fainter, which is why they sound bright, hollow and soft. Subtractive synthesis starts from a rich wave and takes harmonics away.

VCO: THE SOURCE

The voltage-controlled oscillator sets the pitch and the raw timbre. Fourm has two; brighter waveshapes carry more harmonics for the filter to work with.

VCF: THE SCULPTOR

The low-pass filter removes harmonics above the cutoff frequency, darkening the tone as you close it. Resonance emphasizes the region right at the cutoff (module 03).

MIXER: FOUR SOURCES

Osc A, Osc B, the noise generator and the feedback return meet here. If nothing is turned up, nothing reaches the filter: every patch starts at the mixer.

VCA: THE HAND ON THE VOLUME

The voltage-controlled amplifier sets loudness moment by moment, driven by the amp envelope: silence before the key, a shaped rise and fall after it (module 04).

TRY IT From the Basic Program, hold a note and turn OSC A in the mixer all the way down: no source, no sound. Bring it back, then make sure SAW on Oscillator B is lit (press it if it is dark) and raise the OSC B knob in the MIXER: two oscillators, one note, already thicker. With a finger, follow the road across the panel, left to right.

Minimoog players ran a cable from the spare main output back into the external input until the mixer overloaded; the Fourm builds that idea in, with soft-clipping diodes.

OSCILLATORS

Where every sound is born. Each of Fourm's two oscillators offers a set of waveshapes, selected with dedicated buttons and freely combined: Oscillator A has sawtooth and pulse, Oscillator B adds a triangle. The shape you choose decides the harmonic material everything downstream will carve into, and two humble additions, a sub layer and a pinch of noise, separate thin patches from full ones.

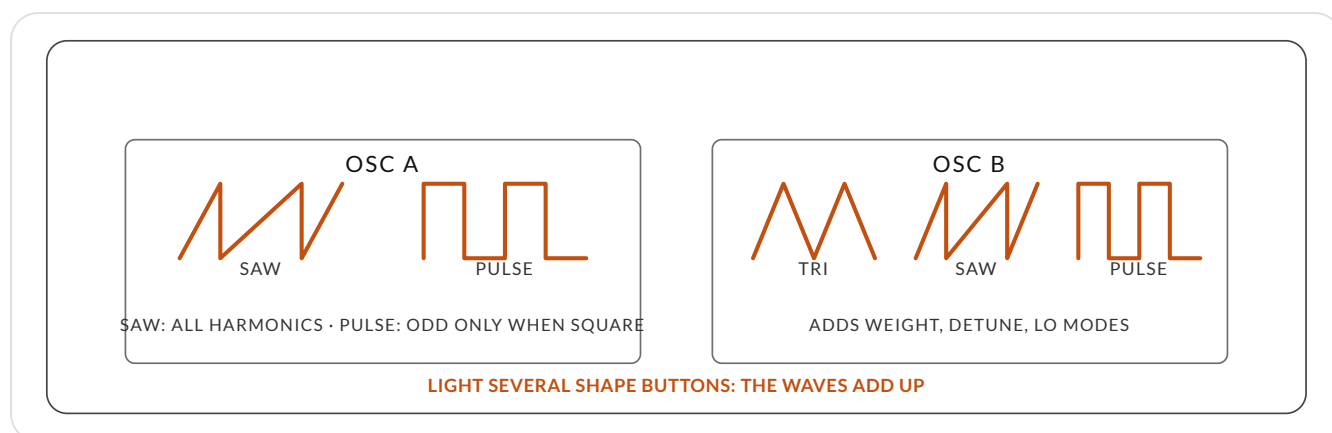


FIG. 2 · THE WAVESHAPES PER OSCILLATOR: OSC A SAW · PULSE, OSC B TRI · SAW · PULSE, LIGHT SEVERAL AND THEY ADD UP.

SAW · TRI · PULSE

Sawtooth carries every harmonic: bright and full, the classic synth backbone, the sound of brass and strings. Triangle is soft and nearly pure, the natural sub layer. Pulse sits in between, hollow like a clarinet at square, thin and reedy as it narrows, and its character depends on its width. A square has no even harmonics, so the octave above the fundamental (the 2nd harmonic) and all its multiples are missing; a clarinet sounds hollow for much the same reason. Move the width off square and the even harmonics creep back in: that is where the reedy character comes from.

OCTAVES & TUNING

Each oscillator has five octave positions, from -2 to +2. Setting A and B an octave apart is the fastest way to a bigger sound. FREQUENCY fine-tunes by up to seven semitones either way; a few cents of detune between A and B makes the two waves beat against each other, and the tone starts to breathe. They drift in and out of step, and their sum swells and dips at the difference between the two frequencies: three cents at 220 Hz is about one swell every three seconds, chorus-like; a few hertz apart and it is a fast throb; past roughly twenty hertz the ear stops hearing a beat at all. Push it too far and you hear two notes: bring it back until it just shimmers.

LEVEL 2

Below the octaves, Osc B has two low modes: LO1 still follows the keyboard, LO2 free-runs. Down there it stops being a voice and becomes a modulation source, like an extra, polyphonic LFO. Sync locks Oscillator A's cycle to Oscillator B for aggressive, tearing timbres. Set A an octave or two above B and turn A's FREQUENCY: the timbre tears, the pitch stays with B (full recipe: LEAD, module 04). Playing alone? Hold PULSE once so the display shows Pulse Width, let go, press SELECT, then turn it with one hand while the other holds a key.

On the Minimoog, in 1970, the third oscillator could be dropped below the keyboard's range and taken out of the mixer, so that it moved the other two instead of sounding: the job Osc B does here in LO1 and LO2.

BUILD IT

PULSE WIDTH & PWM

Pulse width is the duty cycle of the pulse wave, from square to a thin sliver. Fourm has no dedicated knob: hold PULSE and turn SELECT. 0 is the square; turn either way and the pulse thins until it vanishes at the end of the range (the display counts to -63/+64; the negative side gives the same widths with the pulse flipped). Modulating the width with the LFO, pulse width modulation, keeps the tone in constant, chorus-like motion from a single oscillator. We do it in module 05.

THE SUB TRICK, AND NOISE

Set Osc B to triangle, one octave below Osc A: the nearly pure wave slips under the main tone and adds weight without clutter. Noise is the only source with no pitch at all, white (full band), pink (warmer), violet (tilted to the highs). Press NOISE (it comes in at full level, 127) then hold it and turn SELECT down for the level. White is the default; pink and violet live in the PARAM menu: press PARAM, turn SELECT to 4: Noise Type, press SELECT, choose, then press PROGRAM to leave the menu. A trace sharpens attacks and adds breath; more of it becomes percussion.

TRY IT From the Basic Program: on Oscillator A press PULSE so both saw and pulse are lit, then hold PULSE and turn SELECT toward the edge and back. On Oscillator B make sure SAW is lit, leave the octave at 0 and raise OSC B in the mixer; now nudge Osc B's FREQUENCY slightly off centre and listen to the beating (too far and it splits into two notes) then bring it back to the click at 12 o'clock. Press SAW to switch it off, press TRI and set Osc B's octave to -1: add it, take it away, feel the weight go. Press NOISE (it comes in loud), hold it and turn SELECT down to about 30, just a breath of air. Leave it like this: it is the raw material for module 04.

FILTER

Shaping your tone. The oscillators give you a block of harmonics; the filter is the chisel. Fourm's low-pass filter lets everything below a chosen frequency through and fades everything above it: close it and the sound darkens, open it and it brightens. It is the most important knob on any synthesizer, and the one you will touch most tonight.

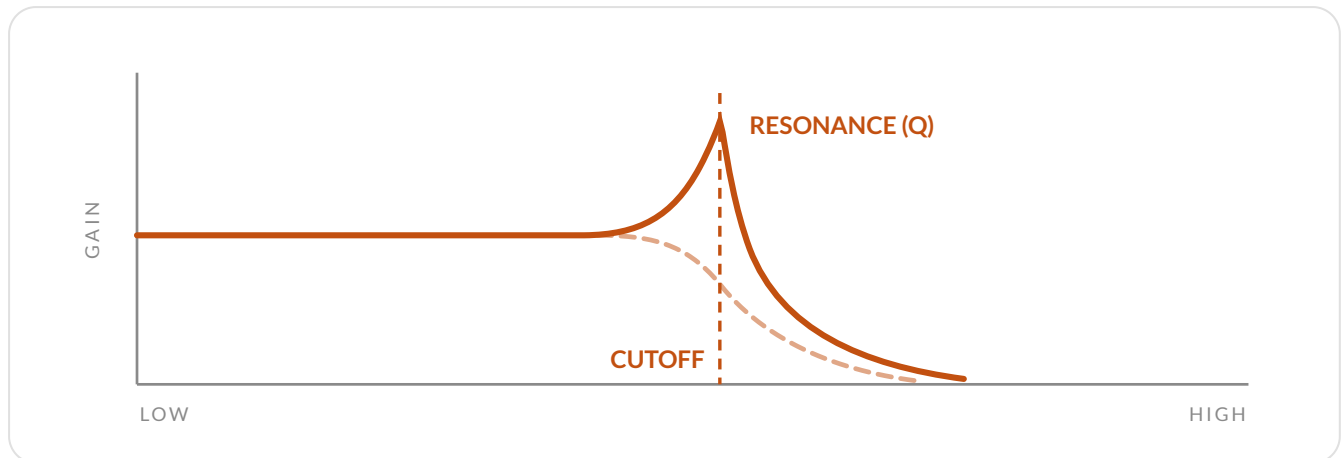


FIG. 3 · A LOW-PASS FILTER WITHOUT AND WITH RESONANCE: THE CUTOFF, AND THE PEAK THAT SINGS.

CUTOFF

The hinge frequency. Below it the harmonics pass; above it they fall away, on Fourm with a steep four-pole slope. Fully open, you hear the raw wave; fully closed, you hear nothing at all, after the mixer, the next thing to check when a patch goes silent.

SELF-OSCILLATION

At maximum resonance the filter feeds on its own output and keeps sounding by itself, a pure sine with no oscillator on at all: a fifth sound source. Turn the volume down a touch before you go there: it arrives suddenly and loud.

WHY IT WHISTLES Resonance feeds a little of the filter's output back to its input at the cutoff, so that band rings the way a string in front of a speaker answers only its own note. Feed back enough and the filter tips over: a microphone too close to a speaker, tuned to the cutoff.

RESONANCE

A boost of the narrow band right at the cutoff. A little adds focus and vowel-like character to a sweep; a lot makes the filter whistle along with the note: that whistle is the resonant peak on its way to self-oscillation. Unlike many classic filters, Fourm keeps the low end full as resonance rises.

KEY TRACK & ENV AMOUNT

KEY TRACK makes the cutoff follow the keyboard: press the button to switch it on (LED lit), hold it and turn SELECT for the amount, quarter, half or full. Higher notes get a brighter filter, as acoustic instruments do; with full tracking, a self-oscillating filter plays a scale in tune. ENV AMT sets how much the filter envelope moves the cutoff over the life of a note, positive or negative, with zero at 12 o'clock. That is the story of the next module.

LEVEL 2

Fourm's filter is a four-pole resonant low-pass in the Prophet-5 lineage. Its bass compensation is always on, so resonance never thins the fundamental the way the original Prophet-5 (and the Moog ladder) do.

Cutoff is not where the filter starts working: the signal is already coming down at the cutoff frequency, and the slope, 24 dB per octave from those four poles, says how fast it falls above that.

Moog's ladder filter, patented in 1969, stacks four stages of 6 dB per octave into the 24 dB slope you are hearing; the patent pushed every other company to invent a filter of its own.

SWEEP IT

TRY IT Hold a chord. Turn CUTOFF slowly down until the sound is dark, then back up. Set RESONANCE to about 3 o'clock and sweep CUTOFF again: a whistle rides the sweep. Lower MAIN a touch, turn RESONANCE to maximum, close CUTOFF a little and play single notes: the filter self-oscillates: it sings on its own, at one fixed pitch. Press KEY TRACK so its LED lights, hold it and turn SELECT to Full, then play a scale: the whistle now follows the keys, in tune. Bring RESONANCE back to 9 o'clock and play low and high: the high notes come out brighter. MAIN back to 12 o'clock, CUTOFF to 12 o'clock, we keep this for the next module.

ENVELOPES

Controlling your sound over time. An oscillator through a filter is still a beep: it starts when you press and stops when you let go, and nothing happens in between. Real instruments have a shape: an organ starts and stops instantly, a plucked string starts loud and fades, a bowed note swells. The envelope is that shape, drawn with four knobs and replayed every time you press a key.

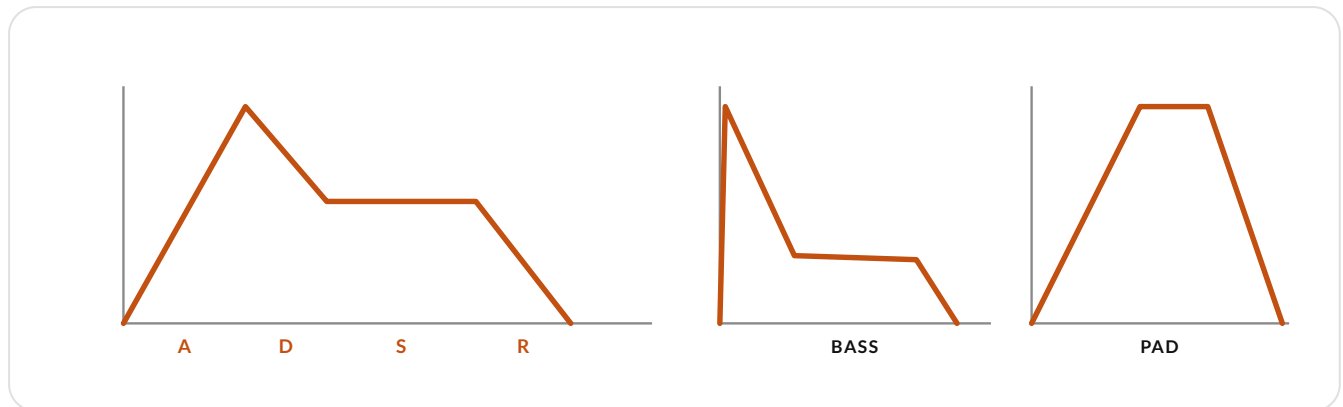


FIG. 4 • THE ADSR ENVELOPE, AND TWO SHAPES: BASS AND PAD.

ADSR

Attack is the time to rise from silence to the peak. Decay is the time to fall from the peak to the sustain level. Sustain is a level, not a time: where the sound sits for as long as the key is held. Release is the time to fade to silence after the key is let go. Your brain identifies an instrument in its first milliseconds: attack matters most.

THE COOKBOOK

Sustain at maximum: the decay has nowhere to fall and does nothing. Sustain at zero: the note dies while you hold it, a pluck. Attack and sustain at zero: percussion. Filter envelope slower than the amp envelope: a pad that opens up after the note has started. The amp is the gate: whatever the filter envelope does while the amp is still closed, you never hear, a fast filter sweep under a slow amp attack, or a long filter release after a short amp release. Filter decay is the pluck knob: shorten it and the sound bites.

The four controls come out of a conversation in 1965: Vladimir Ussachevsky, who ran the Columbia-Princeton Electronic Music Center, asked Bob Moog for a module that would shape a note over time. The first one had its knobs labelled T1, T2, ESUS and T3; the name ADSR came later.

TWO ENVELOPES ON FOURM

The AMP ENVELOPE drives the amplifier: the loudness contour of a note. The FILTER ENVELOPE drives the cutoff, scaled by ENV AMT in the filter section: the brightness contour. Same shape, two jobs. ENV AMT is bipolar: zero at 12 o'clock, positive clockwise, inverted counter-clockwise; an amount left at 12 o'clock is the classic "why is nothing happening".

VELOCITY

Each envelope has a VELOCITY button: switch it on, and how hard you strike scales that envelope's amount (hold it and turn SELECT for the depth; default 127). Velocity to the filter envelope is why soft playing stays dark and hard playing bites (module 07).

SHAPE IT

TRY IT

RECIPE 1 • BASS

Start from where module 03 left you: the module-02 sound (A saw + pulse, B triangle an octave down), NOISE off, RESONANCE at 9 o'clock, KEY TRACK on. Amp envelope: ATTACK 0, DECAY 10 o'clock, SUSTAIN 8 o'clock, RELEASE 8 o'clock. Filter envelope: ATTACK 0, DECAY 10 o'clock, SUSTAIN minimum, RELEASE 8 o'clock; in the filter, ENV AMT to 3 o'clock and CUTOFF to 9 o'clock. Play short, low notes. Now shorten the filter DECAY: more pluck.

RECIPE 3 • PLUCK

From the bass: filter envelope SUSTAIN at minimum, DECAY at 9 o'clock; amp envelope SUSTAIN at minimum, DECAY 10 o'clock, RELEASE 9 o'clock; RESONANCE 10 o'clock. Every note dies by itself, and the shorter the filter DECAY, the sharper the bite.

RECIPE 5 • KICK (NO OSCILLATOR)

From the Basic Program. OSC A to zero in the mixer, RESONANCE to maximum, CUTOFF low, KEY TRACK on and Full, ENV AMT at 3 o'clock; filter envelope ATTACK 0, DECAY 9 o'clock, SUSTAIN 0; amp envelope SUSTAIN 0, DECAY 10 o'clock. The filter's own sine drops in pitch on every note: a kick drum. Higher notes and a longer decay give you a laser.

RECIPE 2 • PAD

Amp envelope: ATTACK 12 o'clock, DECAY 0, SUSTAIN maximum, RELEASE 2 o'clock. Filter envelope: ATTACK 2 o'clock, DECAY 2 o'clock, SUSTAIN 3 o'clock, RELEASE 2 o'clock, slower than the amp, on purpose. CUTOFF to 12 o'clock, ENV AMT to 2 o'clock (right of twelve); on Osc B press SAW (TRI can stay lit), octave 0, OSC B up in the mixer, FREQUENCY nudged slightly for detune. Hold a chord and let it bloom. (Knob positions are starting points, not rules; trust your ears.)

RECIPE 4 • LEAD (HARD SYNC)

From the Basic Program. On Osc A press SAW off and PULSE on; OSC B full up in the mixer; press SYNC on Osc A; Osc A octave +1, Osc B octave -1. Filter envelope ATTACK 1 o'clock, DECAY 1 o'clock. In MODULATION press ROUTE under FILT ENV once (red) with its AMOUNT fully right, then press FREQ A until it is red. Play: the filter envelope sweeps Osc A's pitch against the locked Osc B, the tearing sync lead of "Let's Go".

LFO & MODULATION

Adding movement. Envelopes give a note a shape; modulation keeps it alive. A low frequency oscillator is an oscillator too slow to hear as a pitch: you hear what it moves. Send it to the pitch and you get vibrato, the violinist rocking a finger on the string; to the cutoff, a wah; to the amplifier, tremolo, the volume knob swung twice a second. Modulation is the verb of synthesis: something moves something else.

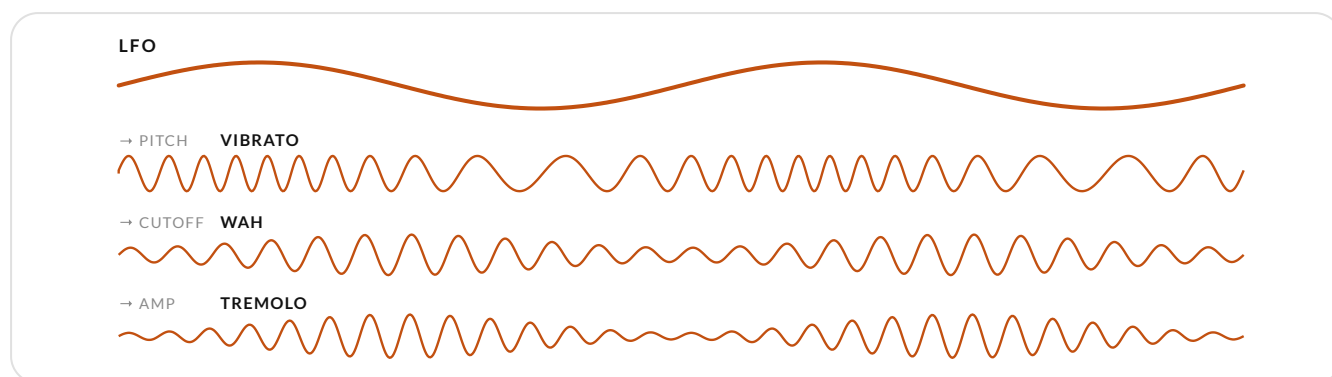


FIG. 5 · ONE LFO, THREE DESTINATIONS: VIBRATO, WAH, TREMOLO.

THE LFO

Its FREQUENCY knob is the speed, from a slow drift to a fast buzz; SYNC locks it to the clock. Its shapes stack: triangle for smooth vibrato and tremolo, sawtooth for ramps, square for trills and on-off tremolo, sample & hold for stepped random motion. Musical vibrato is small and around six to seven cycles per second; speed it up and it stops being vibrato and becomes a growl.

THE CLASSIC MOVES

Vibrato: LFO triangle → FREQ A and FREQ B, a small amount, on the wheel. Wah: LFO → CUTOFF, a sizable amount, at a medium speed, once or twice a second; slow it right down and it becomes a sweep. Tremolo: LFO square or triangle → AMP. PWM: LFO → PW A, the pulse that never stands still.

There is one LFO on the Fourm and this page is the useful half of it. What it shares between the voices, and the rest of the modulation options, are in the official Fourm manual: the QR code on the last page of this handbook.

THE MODULATION SECTION

Three sources on the left, FILT ENV, OSC B and the LFO, each with a ROUTE button and a bipolar AMOUNT knob (zero at 12 o'clock). Eight destinations on the right: FREQ A, FREQ B, PW A, PW B, CUTOFF, AMP, LFO FREQ, LFO AMT. Press ROUTE once for the direct bus (LED red) and the modulation is always on; press it twice for the mod-wheel bus (blue), and it only appears as you raise the wheel: the wheel becomes the volume of the modulation. Destinations work the same way: once for the direct bus, twice for the wheel bus, three times for both (purple).

WIRE IT UP

LEVEL 2

Taken below the octaves, in LO1 or LO2, Osc B stops being a voice and becomes a second LFO, one for each note you hold. In LO1 it follows the keyboard, so high notes wobble faster than low ones. In LO2 it free-runs, at the same speed for every note. Osc B keeps modulating with its mixer level at zero: the mixer sets how loud Osc B is as a sound, while the modulation is taken separately, through the OSC B route. Down at lo1 and lo2 you want it at zero anyway, because as audio it is only a faint clicking. The filter envelope is a modulation source too: FILT ENV → FREQ A gives the “pitch blip” of synth brass; → PW A, a pulse that narrows over the note.

TRY IT On the pad from module 04: in the LFO section make sure TRI is the only lit shape and set FREQUENCY to about 10 o'clock. In the modulation section look at the LEDs: from the Basic Program, ROUTE under LFO, FREQ A and FREQ B are already blue, the mod-wheel bus. If one of them is dark or red, press it until it turns blue. Push the mod wheel all the way up, hold a chord and turn the LFO AMOUNT from 12 o'clock clockwise until you hear the vibrato; bring the wheel down and up again: vibrato that you control. Keep the wheel up. Now press FREQ A and FREQ B until they go dark, press PW A until it is blue: the pulse breathes; then CUTOFF until it is blue: wah. Faster FREQUENCY, more AMOUNT: go too far, then bring it back.

LEVEL 2

The LFO has more than the four shapes you can see. Hold SAW to flip it into a falling ramp; hold S/H|MODE to choose what the S/H button does: sample & hold, Random (a new value on every key, whatever the speed), Pink, White and Violet noise as wobble, and DC, a steady value, which turns the mod wheel into a hand-driven knob for any destination. Only the triangle swings both ways around the note; square, saw and S/H move in one direction only, which is why a square on pitch gives a trill rather than a wobble. With SYNC on, the speed is no longer the FREQUENCY knob but a division of the tempo, chosen in the Param menu (14: LFO Sync Freq). The full range runs from one cycle every 48 seconds to 556 a second, and the top of it is already a tone.

VOICES & UNISON

Four voices, one note, and the loop that thickens everything. Fourm is polyphonic: it holds four complete voices, each with its own oscillators, filter and envelopes, so four notes can sound at once. This module is about how those voices are spent: spread across a chord, stacked on one massive note, or replayed as a stored chord from a single key. And it adds mass from a different place: the feedback loop.

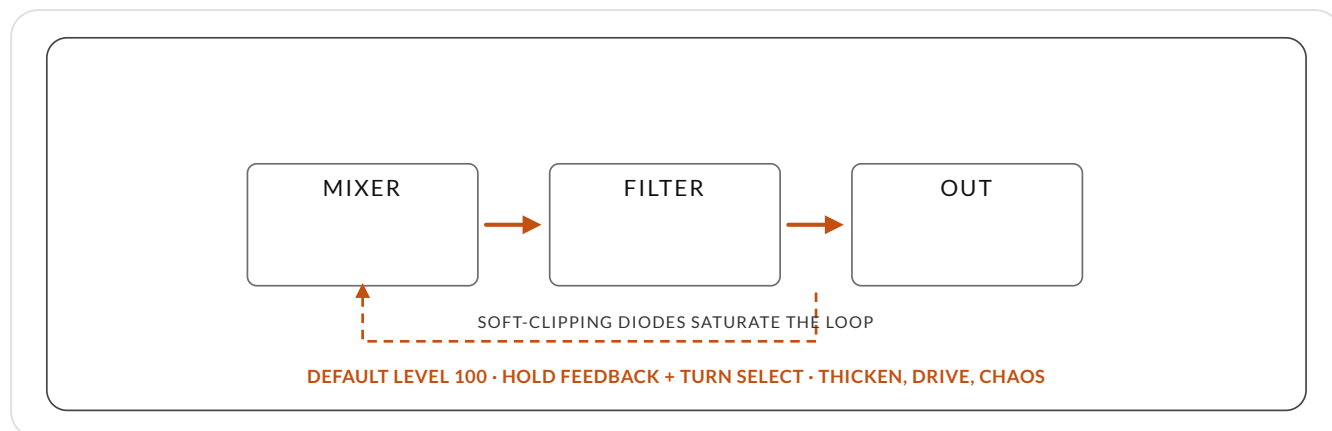


FIG. 6 • FEEDBACK: THE OUTPUT RETURNED INTO THE MIXER THROUGH SOFT CLIPPING.

POLYPHONY

A mono synth plays one note at a time: a second key takes over from the first. Fourm plays four; press a fifth and the oldest note is stolen. With unison on there is nothing left to steal: the synth is monophonic and every new key replaces the last. HOLD sustains up to four held notes after your hands leave the keys, at the amp envelope's SUSTAIN level, so with sustain at zero it holds silence.

CHORD MEMORY

Hold a chord and store it; from then on any single key replays the whole chord, with the notes spaced exactly as you played them, moved up or down as you move along the keyboard. One finger, full harmony, and it is how those short, punchy chord stabs are played live. It is saved with the program.

UNISON

Unison stacks the voices you choose, one to four (hold UNISON and turn SELECT), on a single note. Fourm becomes monophonic and, with all four, enormous. On its own that is just louder; the magic is detune (Param 10, Unison Detune, 0-127), which spreads the voices a few cents apart into one wide, beating tone. Small reads as analog richness, large as an obvious chorus. When it sounds huge, back it off ten percent: that is usually the keeper.

FEEDBACK

A dedicated mixer channel returns Fourm's output into the mix, with soft-clipping diodes in the loop. Press FEEDBACK (it comes in at level 100), hold it and turn SELECT to set the amount. Low amounts warm and thicken; high amounts growl and overdrive, all inside the voice. Unison plus feedback is Fourm's biggest bass.

LEVEL 2

Chord memory, step by step: press UNISON; push SELECT to move to the middle row and scroll to Chord; press UNISON to move to the top row; play and hold a chord of two to four notes; press UNISON again: Chord Mode is on. To clear it: hold UNISON, set the voice count with SELECT, press UNISON off, save. Vintage (Param 11) adds voice-to-voice drift without unison, like an old synth on a warm night.

TRY IT Play five notes at once and count what you hear. Hold UNISON, turn SELECT to 4 voices and play a line: one huge note. Press PARAM, scroll to 10 Unison Detune, press SELECT and turn it to about 40, press PROGRAM to exit: the tone widens. Press FEEDBACK, hold it and turn SELECT down to about 60: growl without chaos. Turn unison off, play a chord, press HOLD (it sits next to UNISON) and lift your hands. Then press HOLD again so the notes let go, and press FEEDBACK once more to switch off the feedback you turned on a moment ago. That way module 07 starts from a clean sound.

The Oberheim Four Voice, 1975: polyphony was four complete monosynths in a single box.



EXPRESSION

Play it, don't just press it. Envelopes and LFOs make every note the same; what makes it music is what your hands add. Velocity reads how fast you strike, aftertouch reads how hard you keep pressing, glide lets one note slide into the next. Fourm's Tactive keybed senses pressure under every single key, polyphonic aftertouch, so each finger shapes its own note.

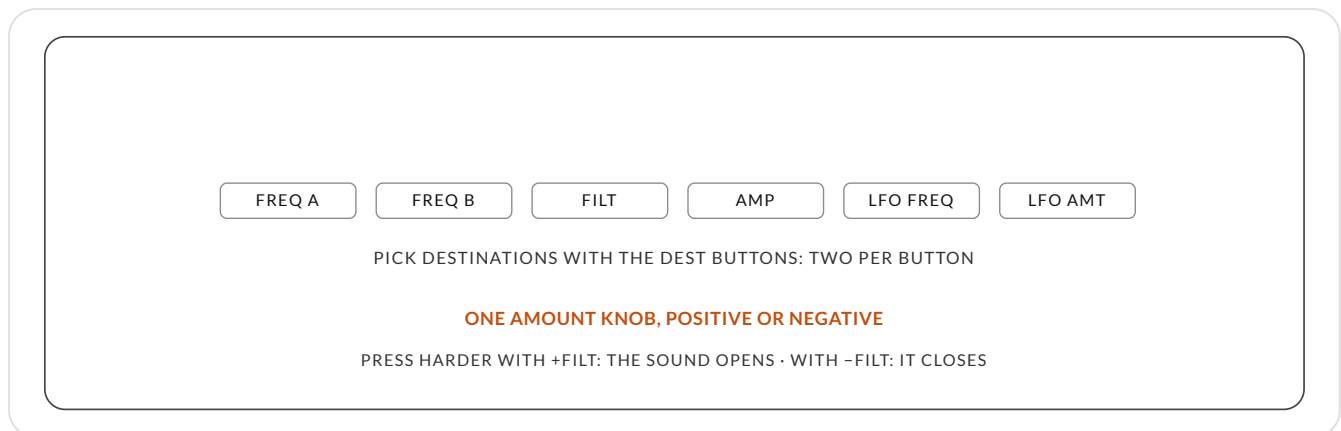


FIG. 7 • PRESSURE ROUTING: ONE AMOUNT, SIX DESTINATIONS.

VELOCITY

Louder notes on real instruments also have more harmonics, which is why velocity belongs on the filter, not just the volume. Each envelope has a VELOCITY button: press it, and how hard you play scales that envelope's depth (hold it and turn SELECT for the amount). Velocity to the filter envelope: soft playing stays dark, hard playing bites. Five response curves in Global, Soft+, Soft, Medium, Hard, Hard+, match the keybed to your touch.

PRESSURE ROUTING

Six destinations on three DEST buttons, two on each: FREQ A and FREQ B, FILT and AMP, LFO FREQ and LFO AMT. Above every button sit two small LEDs, one for each of its pair, and they tell you where the pressure is going. Press once and the left one lights, the first destination. Press again and the right one lights instead, the second. Press a third time and both light, both destinations at once. A fourth press turns them off. One AMOUNT knob serves all of them, positive or negative, with zero at 12 o'clock, like ENV AMT. Cutoff is the classic: press for brightness, release to relax; a negative amount closes the filter instead. Pitch is for the brave.

Sliding between notes is older than synthesizers: on the Ondes Martenot, in 1928, the right hand wore a ring on a ribbon, and the pitch moved continuously.

POLY AFTERTOUCH

After the key is down, extra pressure becomes a continuous modulator, per key. Lean into one note of a held chord and only that note blooms, like blowing harder into a flute: press harder, brighter. Mono aftertouch, on most keyboards, moves every voice at once; per-note aftertouch used to be a rarity, the CS-80, the Prophet T8, and Fourm has it natively. Turn it on in Global 9 Aftertouch, set to Poly; Global 10 Atouch Curve gives it five curves of its own, Soft 2, Soft 1, Medium, Hard 1, Hard 2, and Medium is the linear one.

GLIDE

Also called portamento: the pitch slides from the previous note to the next instead of jumping. Press GLIDE and set RATE: glide on with rate at zero does nothing. Subtle settings, barely a slur, sound the most musical: it is the difference between a keyboard melody and a phrase that sounds like a voice.

HANDS ON

The settings behind what you just played, and then the exercise. Take the exercise slowly: every step is one hand on one control, and the point is to hear the change, not to get through the list.

LEVEL 2

Four glide modes, from GLIDE + SELECT or from Param 17. The A stands for Auto: the glide happens by itself, only when two notes overlap, that is, only when you hold one note until you press the next.

- **Fixed Rate:** the default. The rate is fixed, so a bigger jump takes longer.
- **Fixed Rate A:** the same, but it glides only when you play legato.
- **Fixed Time:** every jump takes the same time, however big it is.
- **Fixed Time A:** the same, legato only.

TRY IT If HOLD is still lit from the last exercise, press it to let the notes go, and set the filter's ENV AMT to about 2 o'clock. In the FILTER ENVELOPE press VELOCITY so it lights; play soft, then hard. In AFTERTOUCH press the DEST button under FILT / AMP until only FILT is lit and turn AMOUNT from 12 o'clock to 2 o'clock; hold a three-note chord and press into just the top note. Now turn AMOUNT to 10 o'clock: pressure closes the filter instead. Hold UNISON and turn SELECT to 1 voice, press GLIDE, set RATE to 10 o'clock and play a bass line one note at a time: the notes slide into each other. Then press the FILT / AMP button until it goes dark, press the FREQ A / FREQ B button until both are lit, and bring AMOUNT back to just right of 12 o'clock. Now pressing harder bends the pitch upward. The keyboard senses each key on its own, so only the note you lean on bends; the others stay where they are.

ARPEGGIATOR & SEQUENCER

Patterns from held notes. Hold a chord and the arpeggiator plays it back one note at a time, locked to the clock. It turns sustained harmony into rhythm, and Fourm hides a 64-step sequencer behind it. The mouse is not a musical instrument; this is why the arpeggiator lives on the panel.

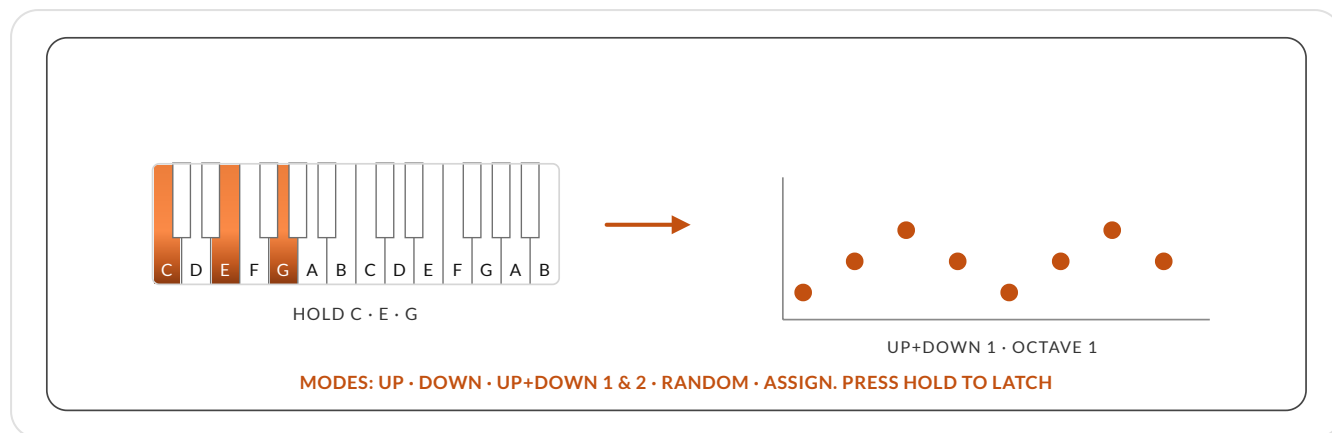


FIG. 8 • HOLD A CHORD, THE SYNTH PLAYS THE PATTERN.

DIRECTIONS & OCTAVES

Press ON in the arpeggiator and hold some notes. Six modes, Up, Down, Up+Down 1 and 2, Random, Assign (as played), over one to three octaves (hold MODE or OCTAVE and turn SELECT). HOLD keeps the pattern running while both hands leave the keys; play a different number of notes and the rhythm changes: it sounds like a mistake until you embrace it.

LIVE MOVES

While a pattern runs, sweep the cutoff, lean into aftertouch, flip divisions. The arpeggiator supplies the rhythm; you play the tone.

CLOCK & DIVISIONS

One clock rules the tempo: tap TAP TEMPO four times, or hold it and turn SELECT for a BPM. CLK DIV (hold it and turn SELECT) picks the rhythmic grid, quarters, eighths, sixteenths, dotted, triplets, and the LFO's SYNC follows the same clock. If the arp will not start, check Clock Mode in Global: set to In, it waits for an external clock.

NOTE & MOD SEQ (LEVEL 2)

Beyond arpeggios, Fourm records up to 64 steps of notes, chords, rests, ties, even a glide per step, triggered and transposed from the keyboard, and a mod sequence can step a parameter instead: cutoff, pulse width, feedback. Each sequencer track is tied to a voice; put chords on a step for polyphony. A sequencer in disguise, and a story for another workshop.

TRY IT Press ON in the arpeggiator, hold C–E–G, press HOLD (in the UNISON section) and lift your hands. Tap TAP TEMPO four times to your tempo. Change CLK DIV to sixteenths (hold it and turn SELECT), then to eighths, hear the grid change. Ride CUTOFF while the pattern runs. Hold MODE and turn SELECT to Random; hold OCTAVE and turn SELECT to 2. Now hold a “wrong” chord and let it repeat: give it eight bars.

A step sequencer began as a row of eight knobs walked through by a clock; that is how the bass of “I Feel Love” was made, on a Moog modular.

WHAT IS IN A SOUND

This one is for later, once the panel feels familiar. Every note has a **fundamental frequency**, the lowest one in the sound, and the one you hear as the note's pitch. Above it sits a series of fainter tones at whole-number multiples of it: twice the frequency, three times, and so on. Those are the **harmonics**, and which ones are present, and how loud, is what makes one waveshape bright and another hollow.

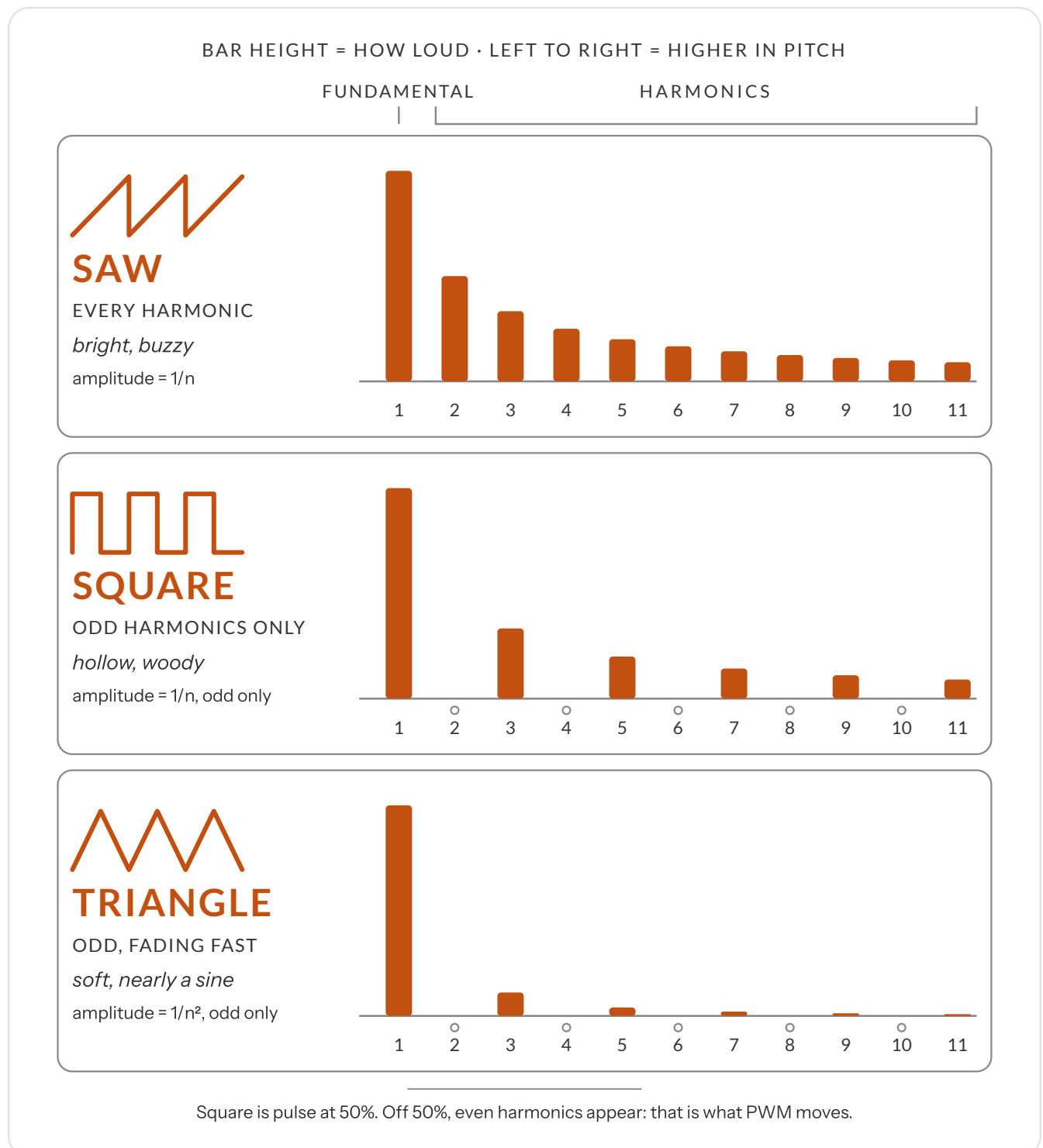


FIG. 10 • THE HARMONIC CONTENT OF THE THREE WAVESHAPES.

FUNDAMENTALS

SIGNAL FLOW

The road a sound takes: oscillators → mixer → filter → amplifier → output.

HARMONICS

The whole-number multiples of a note's fundamental, stacked above it and usually fainter; their mix is what you hear as timbre.

SAWTOOTH

Every harmonic, falling at $1/n$: bright and full, the classic synth backbone.

TRIANGLE

Odd harmonics that fade fast: soft, nearly a sine, the natural sub layer.

OCTAVE

Five positions per oscillator, -2 to +2; FREQUENCY fine-tunes up to seven semitones.

NOISE

Unpitched sound in white, pink or violet: air, breath and percussion.

VCF • CUTOFF

Voltage-controlled filter; cutoff is where the low-pass starts removing highs.

SELF-OSCILLATION

At maximum resonance the filter feeds on its own output and keeps sounding by itself, a pure sine at the cutoff frequency, with no oscillator on. The whistle you hear just before is that peak about to tip over.

ENV AMOUNT

How much the filter envelope moves the cutoff; bipolar, zero at 12 o'clock.

VCA

Voltage-controlled amplifier: the loudness stage, shaped by the amp envelope.

LO1 • LO2

Osc B below the octaves: a second modulator, one per voice, key-tracked (LO1) or free (LO2).

BASIC PROGRAM

The reset sound: hold TRANSPOSE DOWN and press PROGRAM.

PATCH • PRESET

A patch is a sound as the panel is set right now, before you save it; a preset is one already stored in the synth, ready to load.

FUNDAMENTAL

The lowest frequency in a note, and the one you hear as its pitch. Everything else in the sound sits above it.

VCO • OSCILLATOR

Voltage-controlled oscillator: generates the raw waveform. Fourm has two.

SQUARE

Pulse at 50% width: odd harmonics only, hollow and woody.

PULSE WIDTH • PWM

The duty cycle of the pulse wave (hold PULSE, turn SELECT); PWM is modulating it.

MIXER

Sets the level of each source: Osc A, Osc B, noise, feedback.

SUB OSCILLATOR

A layer one octave below the main pitch; on Fourm, Osc B triangle does the job.

RESONANCE

A boost of a narrow band right around the cutoff, not a single frequency; high settings whistle, and at maximum the filter self-oscillates.

KEY TRACK

The cutoff follows the keyboard: off, quarter, half or full.

ENVELOPE • ADSR

Attack, Decay, Sustain, Release. Sustain is a level, not a time.

SYNC

Locks Oscillator A's cycle to Oscillator B: hard, tearing timbres.

SOFT CLIPPING

Gentle rounding of the peaks: saturation instead of harsh distortion.

PARAM • GLOBAL

PARAM is the menu of settings that have no knob (press PARAM, turn SELECT); GLOBAL holds the settings that apply to every sound.

TIMBRE

The colour of a sound: what makes a violin and a trumpet sound different on the same note.

PERFORMANCE & MOTION

LFO

A low frequency oscillator, too slow to hear as pitch: you hear what it moves.

VIBRATO

LFO to pitch.

WAH

LFO to the cutoff, the third of the trio; named after the pedal, not a formal term. Triangle LFO, slow, into CUTOFF with RESONANCE at 2 o'clock: the filter opens and closes on its own, the sound behind a thousand funk records.

MOD WHEEL

The volume of the modulation on the wheel bus.

UNISON

One to four voices stacked on a single note; the synth becomes monophonic.

CHORD MEMORY

A stored chord replayed from a single key, transposed as you move.

FEEDBACK

The output fed back into the mixer through soft-clipping diodes: warmth, then growl.

POLY AFTERTOUCH

Pressure sensed per key by the Tactive keybed; each finger modulates its own note.

GLIDE • PORTAMENTO

The pitch slides between notes instead of jumping; four modes.

CLOCK • BPM • DIVISION

The shared tempo and the rhythmic grid the arpeggiator and sequences follow.

ARPEGGIATOR

Plays the notes of a held chord back one at a time, locked to the clock; HOLD latches it.

RATE • FREQUENCY

The LFO's speed (the FREQUENCY knob on Fourm); SYNC locks it to the clock.

TREMOLO

LFO to amplitude.

MODULATION BUS

Direct (red LED) is always on; mod wheel (blue) appears as you raise the wheel.

POLYPHONY • VOICE

Fourm holds four complete voices; a fifth note steals the oldest.

DETUNE

Small pitch offsets between stacked voices, measured in cents; beating equals width.

HOLD

Sustains up to four notes after your hands leave the keys, at the amp envelope's sustain level; also latches the arpeggiator.

VELOCITY

How fast the key goes down; five curves, from Soft+ to Hard+.

PRESSURE ROUTING

Six destinations on three DEST buttons, with one bipolar AMOUNT knob.

LEGATO

Overlapping notes; in the legato glide modes, only these slide.

NOTE & MOD SEQ

Up to 64 recorded steps of notes, or a stepped lane that moves a parameter.

REST • TIE

A silent step; a step that extends the note before it.

FREQUENTLY ASKED QUESTIONS

Eight questions that come back at the end of every workshop. The short answers are here; the long answer is the panel in front of you.

WHY DO I HEAR NOTHING?

Recall the Basic Program first: hold TRANSPOSE DOWN and press PROGRAM. If the sound comes back, the fault was in your patch, usually amp envelope sustain at zero, cutoff fully closed, or every mixer level down. If the Basic Program is silent too, work through the checks in order on page 20.

WHY DOES A KNOB DO NOTHING?

POT MODE, in the GLOBAL menu. The lined knobs are pots with limited travel, so the synth must decide what to do when the knob position disagrees with the stored value. In PASSTHRU nothing moves until you turn past that value. In RELATIVE the value follows your hand, but the far end stays out of reach until the knob reaches its own end. In JUMP it snaps at once.

WHY IS FOURM OUT OF TUNE?

Check COARSE TUNE and FINE TUNE in the GLOBAL menu; both should read 0. If they do, run the calibration: press GLOBAL, scroll to TUNE VOICES, press SELECT, then INC/SAVE, and let it finish without cutting the power. It is analog: the manual asks you to repeat this over the first few days, while the instrument learns its range of working temperatures.

IS ANALOG DIFFERENT FROM A PLUG-IN?

The theory is identical: same oscillators, same filter, same envelopes, same road from left to right. A plug-in computes those stages; Fourm builds them from circuits whose voltages drift a little with heat and age, which is what people mean by analog. The real difference is the panel, one knob per job, no menus, and that changes how you learn.

FILTER ENV OR AMP ENV?

The same four knobs, attack, decay, sustain, release, moving two different things. The amp envelope shapes loudness: whether the note appears at once or swells, and how long it rings after you let go. The filter envelope shapes brightness: it moves the cutoff. On Fourm the filter envelope is also a modulation source, so it can drive other destinations as well.

HOW DO I SAVE A SOUND?

Hold PROGRAM until its LED blinks, then let go. The display reads WRITE: Program X; press INC/SAVE and the sound is written there. For another slot, turn SELECT while the LED blinks to pick a location from 1 to 128, press SELECT again to change the bank row, then INC/SAVE. Edited factory sounds always land in the User bank. DEC/EXIT cancels.

WHAT DOES THE MOD WHEEL DO?

Nothing fixed. The wheel is a bus, not a function: it raises only what that program routed onto it. In the MODULATION section the blue LEDs show which source sits on the wheel and which destination receives it, and the AMOUNT knob of that source sets how far the wheel can push. The guide warns it can do more than add vibrato, so read the LEDs.

WHICH SYNTH SHOULD I BUY FIRST?

The one whose panel you understand. A synth you can read is a synth you will use; a larger one you cannot read will sit in the corner. You know this road now, oscillators, mixer, filter, envelopes, modulation, and it is the same road on almost every instrument. Judge the next one by how plainly its panel says so.

TROUBLESHOOTING

This page is for home, when there is nobody beside you to ask. Almost everything that feels broken on an analog synth turns out to be one of the things below.

IF YOU HEAR NOTHING

Take them in this order, it follows the signal through the synth.

1. Are the headphones pushed all the way into the splitter, and the splitter into the Fourm?
2. Is MAIN, the volume knob, turned up? At zero nothing comes out, however well the rest is set.
3. In the MIXER, is at least one source up? The four are OSC A, OSC B, NOISE and FEEDBACK. With all of them at zero the filter has nothing to filter.
4. Is CUTOFF closed all the way? Fully closed means the filter lets nothing through. Turn it to the right and listen.
5. Is HOLD still lit from the exercise before? Then the synth is holding the old notes and ignoring what you play now. Press it to switch it off.
6. Is the mod wheel down while the sound depends on it? A blue route in MODULATION listens to the wheel. Push the wheel up.
7. Still nothing? Go back to the starting sound: hold TRANPOSE DOWN and press PROGRAM.
8. Still nothing after that? In the Global menu, 3, Local Control, set it to On. On Off the keyboard stops talking to the synth.

LEVEL 2 • WHEN A KNOB DOES NOTHING

The Fourm remembers a value for every knob in the program. Global 4, Pot Mode, decides what happens when the knob's position and the stored value disagree. Three settings:

- **Jump** — the value follows the knob the moment you touch it.
- **Passthru** — nothing changes until the knob passes through the stored value.
- **Relative** — the knob moves the value from where it was, so you may have to sweep to an end stop to reach the extremes.

In the workshop every synth is on Jump, so “CUTOFF to 9 o'clock” means exactly that. On your own Fourm, if a knob feels dead, check Global 4 first.

IF YOU HEAR SOUND BUT NOTHING CHANGES

These all look the same on the panel, and all of them make a sound feel stuck.

- An amount knob left at 12 o'clock means zero, not half. It happens on ENV AMT, on the three MODULATION amounts and on the AFTERTOUCH amount. Turn from twelve toward the right.
- A destination that is not lit receives nothing, however far you turn the amount.
- KEY TRACK, VELOCITY and GLIDE have an on button as well as a value. With the LED dark the value does not matter.
- The ATTACK knob does not change the note already sounding, only the next one. Press a fresh key to hear it.
- A knob that feels completely dead: in the Global menu, 4, Pot Mode. If it is not on Jump, the knob does nothing until it reaches the stored value.

TO FIX A SOUND

Sequential's own order, from the source toward the output.

1. The oscillators. Right waveshape? Right octave? One or two of them?
2. The filter. Too harsh, bring CUTOFF down. Too dull, bring it up. Look at RESONANCE too.
3. The envelopes. Starts too abruptly, raise ATTACK. Falls too fast or too slowly, work on DECAY and SUSTAIN. Hangs on after you lift your hand, shorten RELEASE.
4. The LFO, last, for anything that should move on its own.

TAKE IT HOME

That is the whole course. Keep this handbook next to the synth: read one module, turn one knob, listen. The order on the panel is the order in your head now, and every synthesizer you meet from here on is the same road with different clothes.

When you build a sound you like, write it down: the waveshapes, the octaves, the filter, the four knobs of each envelope, what the LFO moves. A patch on paper survives; a patch in memory rarely does.

A PUZZLE

“My bass sound is too buzzy, and it fades in slowly. What should I do?” Work it out before you read on: the buzz is harmonics, so that is the filter (module 03); the slow fade-in is time, so that is an envelope (module 04).

Answer: amp ATTACK to zero, then CUTOFF down until the buzz is gone.



SINTEZAUR • TELEGRAM

the community, the gear talk, and where to say hello



SINTEZAUR • INSTAGRAM

what we are building next, in pictures



SEQUENTIAL • FOURM

the user guide, the preset list and the full specs

ZEEDO where to try and buy the Fourm: zeedoshop.ro

SEQUENTIAL the instruments that made tonight possible: sequential.com

WHEN YOU SAVE ON YOUR OWN FOURM

The steps are in the FAQ, page 19. What does not fit there: while the LED blinks you can press GLOBAL|NAME to name the sound, picking each character with SELECT, then INC|SAVE; DEC|EXIT cancels. The two User banks are yours to overwrite and the two Factory banks never change, while User 2 ships as 128 copies of the Basic Program, 128 empty pages. And if you want to forget presets altogether: Global 25, Preset Mode Off, is live-panel mode, where the knobs are the sound. It is the best way to learn.

Thank you for playing.

